

Sri Sathya Sai Institute of Higher Learning

Brindavan Campus



Master Project Team 7

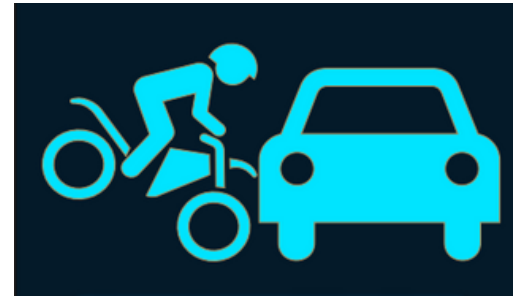
**Marketing
Report**

GUARDIAN X

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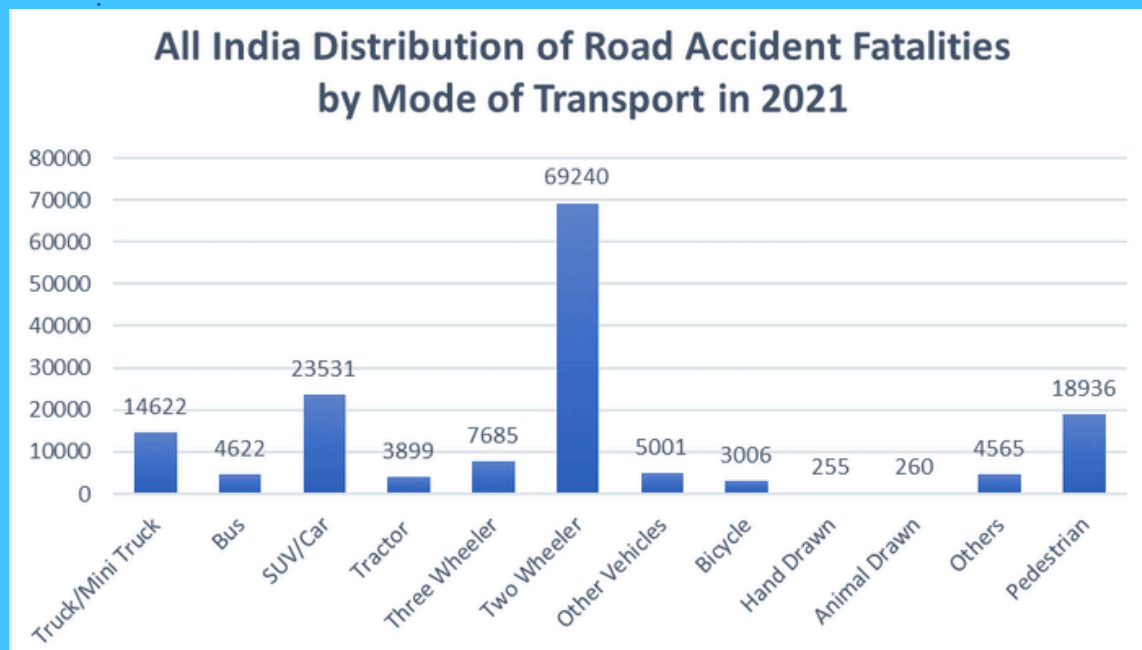
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Problem Statement



What is the problem identified?

Around 93% of road fatalities are concentrated in low and middle income countries like India. According to a report by 'frontline', more than two deaths are occurring per minutes due to road traffic crashes, with over 3,200 per day. More than half of it is faced by two-wheelers. The current generation of youth has low awareness on safety and high need for style.



It could be seen from the graph that in 2021, two wheeler accidents were the highest in Road fatalities.

in 2022, close to 4 lakh accidents occurred in India, with two wheelers category taking nearly 53% of it, leading close to 1 lakh deaths or fatal injuries.

One of the main reasons for this is late emergency response systems and poor helmet conditions offering them high safety.

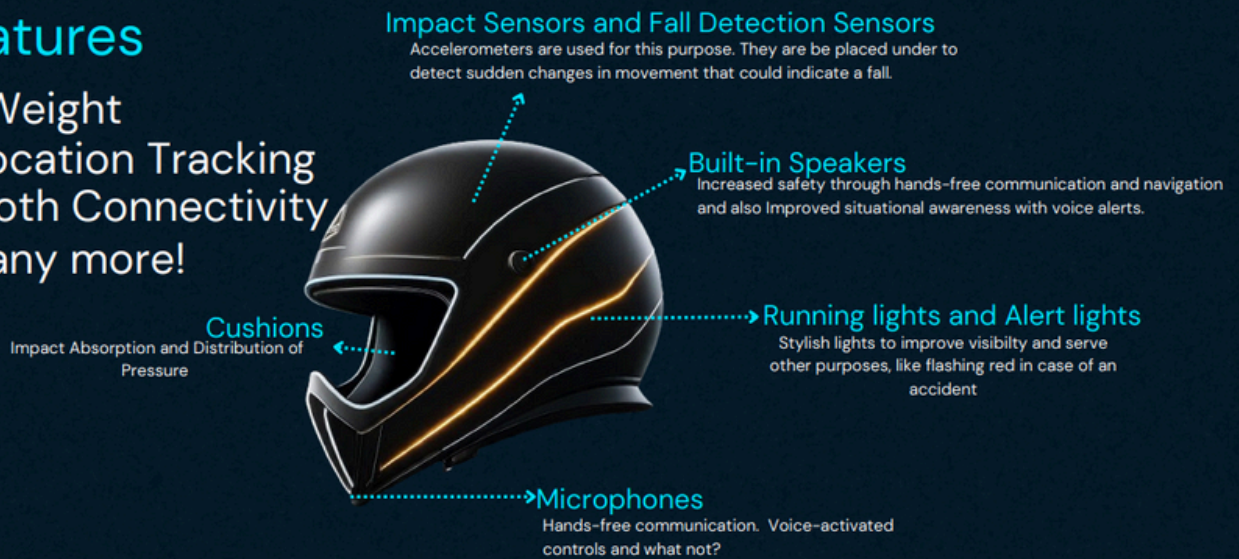
Solution: GUARDIANX

Overview of the Helmet

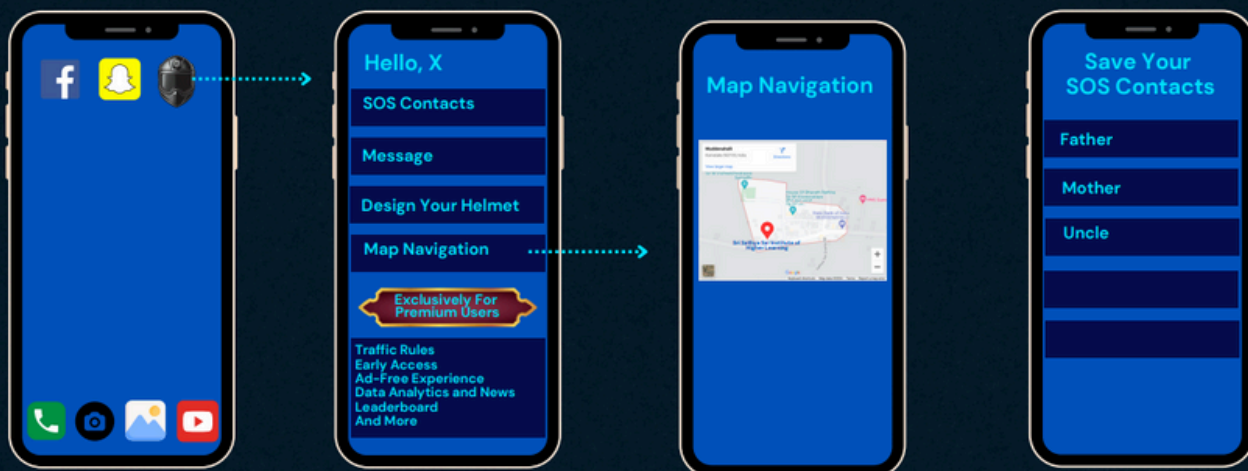


Other Features

- Light Weight
- SOS Location Tracking
- Bluetooth Connectivity and many more!



Mobile Application



Vision

We aim to revolutionize road safety by creating an intelligent helmet that not only protects but also responds actively, offering instant emergency response, seamless communication with comfort for every rider, every ride.

Mission

Our mission is to humanize safety gear by designing and delivering innovative IoT-enabled Safety helmets that protect riders and actively respond during emergencies. We aim to address the critical issue of road fatalities in India by combining impact detection, seamless communication, and superior comfort for riders.

PRODUCT	INDUSTRY	GEOGRAPHY
Develop high quality, technologically advanced helmets that enhance safety and comfort for two wheeler riders.	We aim to lead auto safety gear industry by integrating IoT technology with real-time response systems.	we focus ourselves on Indian geography due to its largest two-wheeler markets globally, addressing the issue of road fatalities.

VERTICAL INTEGRATION	MARKET	COMPETENCE
We aim to collaborate with emergency service providers and telecom networks to enable faster response systems.	We cater ourselves to the safety-conscious, tech-savvy segment for two wheeler riders..	Leveraging our expertise in IoT, wearable tech, and user friendly design to create helmets that are both protective and intelligent combining safety and style.

SWOT ANALYSIS

Strengths

- **Innovative Technology:** IoT-enabled sensors, impact detection, GPS tracking, and emergency response systems provide a unique and safe travelling experience.
- **Life-Saving Capability:** Automatic alert systems for emergency services and loved ones, addressing a critical gap in road safety.
- **Comfort and Functionality:** Our product combines safety with comfort, and has Bluetooth connectivity and hands-free communication.
- **First Mover Advantage:** There are very few helmets that offer integrated impact detection and response systems.

Weaknesses

- **High Production Costs:** Advanced Technology increases variable costs, potentially increasing prices.
- **Possibility of Limited Market Penetration:** High initial costs may limit affordability for the lower-income segment population of India.
- **Reliance on Technology:** We depend exclusively on technology, IoT sensors, and GPS networks, which may pose a challenge in terms of cyber security strong tech teams, and nationwide connectivity.

Opportunities

- **Increasing Road Safety Awareness:** There is a nationwide awareness about the need for road safety and measures are being taken by the GOI to promote the same.
- **Collaboration with two-wheeler manufacturers:** Partnering with two-wheeler manufacturers such as Yamaha
- **Government Regulations:** Support for road safety initiatives and mandatory helmet laws.
- Growth in two-wheeler ownerships (especailly in the EV segment)

Threats

- **Competition from established brands:** Competitors may enter into the field by introducing similar technologies who can grow faster due to Economies of Scale and brand affinity.
- **Technological Failures:** Malfunctions in sensors or connectivity issues could lead to negative user experience that could risk the life of the user.
- **Consumer Skepticism:** Some consumers might not believe in the helmet's features

PESTLE ANALYSIS

POLITICAL

- The Indian Government is promoting road safety through stricter laws for helmet usage.
- With “Make in India” scheme, there could be possible incentives for manufacturing the helmets locally.
- Import duties on electronic costs might increase the production cost thereby impacting pricing of the helmet.



ECONOMIC

- India is experiencing slow economic Growth with rising middle-class income, thereby leading to more demand for premium safety gear.
- Fluctuations in raw material costs, particularly for advanced tech equipment could affect the pricing of the product.
- Being price sensitive, the Indian market could show resistance initially to the purchase of the helmet.

SOCIAL

- Growing awareness about road safety especially due to high accidents rates involving two-wheelers could have an increased demand for advanced safety gear.
- Urbanization, which is on the rise, contributes to a higher demand for two-wheelers and therefore contributes to higher demand for helmet gears.
- Some parts of India may perceive usage of helmets as unnecessary or uncomfortable, and therefore consumer education is required in this area.

TECHNOLOGICAL

- Advancement in IoT enhances the capabilities of the helmet and improve user experience.
- Penetration of 4G and 5G networks facilitates reliable emergency alerts and GPS services that could be more accurate.

LEGAL

- Compliance in the areas of IoT device certifications and ISO-certified helmets is necessary to enable hassle-free production and sales.
- Privacy Laws could pose a challenge as collection of the rider's driving pattern is required for improvement which could involve collection of data.

ENVIRONMENTAL

- Customers' demand for eco-friendly materials in the products they use necessitates the need for the helmet to be sustainable and eco-friendly.
- There is a requirement of recycling the e-waste once disposed and find recycling usage to reduce negative environmental impact.
- Energy-efficient batteries in the helmet are required to reduce the disposal of batteries to minimize the after-effects on environmental sustainability.

PRODUCT CONCEPT

“The Life-Saving Smart Helmet is an advanced protective gear designed specifically for two-wheeler enthusiasts & urban commuters who prioritize safety and convenience. This smart helmet integrates emergency response systems, impact detection, and GPS tracking, ensuring immediate assistance in the event of an accident. With hands-free Bluetooth communication and a design tailored for prolonged comfort, it caters to daily riders navigating heavy city traffic, by offering seamless technology integration and superior safety, the helmet transforms daily commutes into secure, connected experiences.”

Based on the product concept, the product is mainly targeted as follows:

Customer Group: Two-wheeler & Travel Enthusiasts, Long Distance Commuters, Urban Commuters (predominantly two - wheeler)

Usage: The product can be used just like a normal helmet, whenever the customer is taking his two-wheeler out of the parking space.

Benefits: The helmet offers immediate emergency response systems, impact detections, GPS tracking, to assist the rider in the event of any accident, big or small.

MARKET RESEARCH

To understand the behavior of the customer for this product concept, a questionnaire was prepared and sent covering the following areas:

- Intended Purchase Measures
- Overall Product Diagnosis
- Special Product Attributes
- Respondent Profiling Variables

Here is a brief snippet of the questions asked (in order):

- Respondent Profiling Variables:
 - Personal Details such as Name, email ID, Contact No.; City and State of Residence - these were kept open ended
 - Occupation (Student/ Working Professional / Other)
 - Age Group
 - Whether the respondent has a two wheeler?
 - From where does the respondent purchase his helmet - Online Stores/ Physical Store Outlets/ Other (this option was kept open ended)
 - How much does the respondent typically spend on a helmet?
 - Who influences their purchase decision?
 - How would they rate the level of safety offered by their current helmet?
- Intended Purchase Measures:
 - How likely is the respondent interested in buying this product?
 - What Price are they willing to pay for it?

- Overall Product Diagnostics
 - Uniqueness of the product
 - Believability on part of respondents
 - Importance in solving their problems
- Specific Product Diagnostics
 - Respondents' perception of the performance of the helmet
 - Importance of product's features for the respondent

A simple random sampling method was adopted to collect responses. The google forms covering the questions was kept open for 2 days and the total responses gathered over the 2-day period is 123 responses. The form was sent in such a way that there could be equal possibility of various age groups responding to the questionnaire since the product is aimed to be used by all age groups. Below shown are graphs for some of the questions asked from the respondents.

Based on the information, how likely are you interested in buying this product?

123 responses

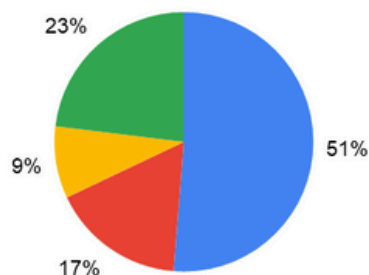


Based on the information, how likely are you interested in buying this product?

Count of Based on the information, how likely are you interested in buying this product?

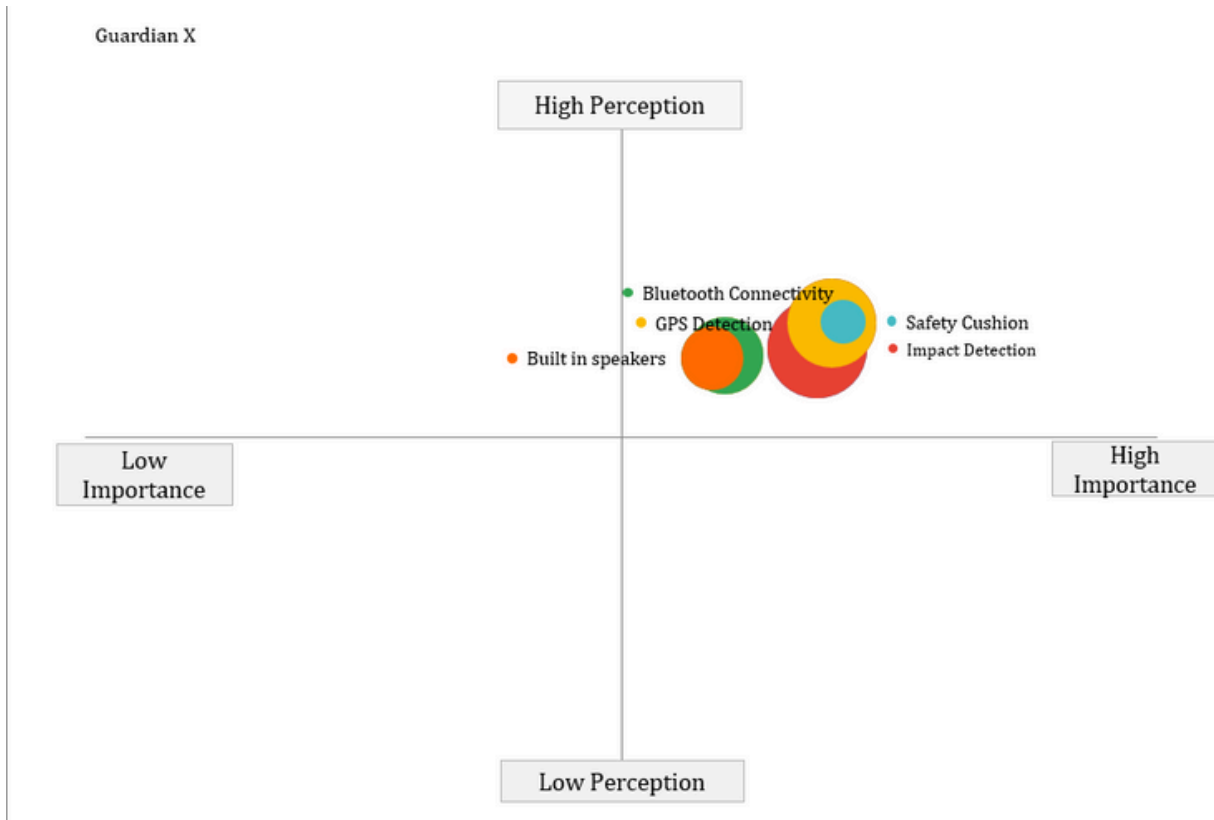
Total

How old are you?



18 - 24 25 - 34 35 - 44 45 +

These two charts shows how age factor for our product play a major role. Our target audience being two-wheeler enthusiasts, clearly shows that the age group between 18-24 are the ones who majorly want to buy the product. Secondly being dominant by the adult age group who state the need or the importance of transmission from one place to another and not only the two-wheeler enthusiasts.

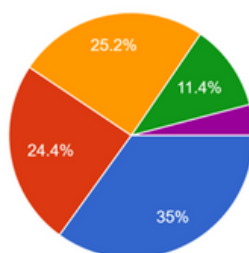


Importance-Perception Matrix

From the matrix above, it could be seen that the respondents have indicated that the safety features offered by the helmet are highly important and they perceive it to be fully capable of performing it. Therefore, this indicates that the concept was well received by the respondents.

How would you rate the level of safety provided by your current helmet on a scale of 1 to 5?

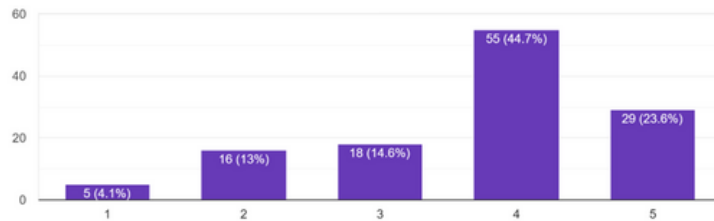
123 responses



- 1 - I am very confident that my helmet will give the maximum protection
- 2 - I am confident that my helmet is better than all other helmets
- 3 - I am confident that my helmet has some protective features
- 4 - I am unsure whether my helmet has protective features
- 5 - I am not at all confident about my helmet and am planning to buy a new...

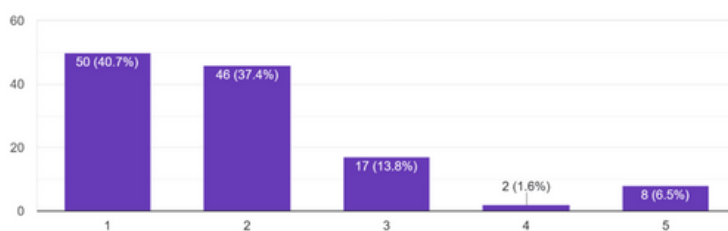
Uniqueness (How different is this helmet compared to helmets currently in the market?) 1 - Very Similar 2 - Similar 3 - Neutral 4 - Somewhat different 5 - Very Different

123 responses



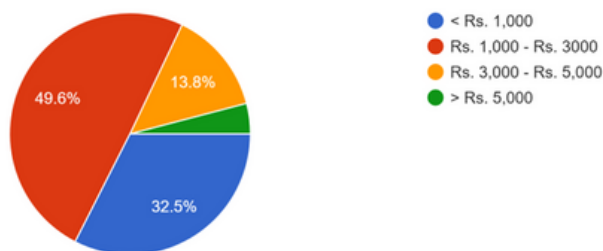
Problem Solving (How important is the helmet in solving your problem?) 1 - Very Important - The helmet effectively addresses my safety concerns. 2 ... my safety concerns is not a buying factor for me.

123 responses



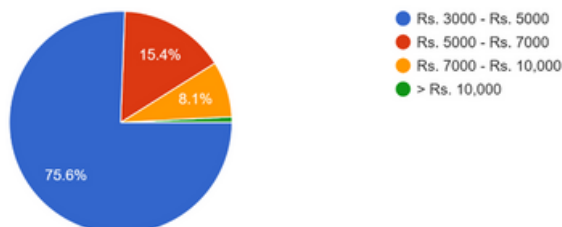
How much do you typically spend on a helmet?

123 responses



ONE FINAL QUESTION How much are you willing to pay for this helmet equipped with such safety features?

123 responses



The two graphs to the left shows that the respondents feel the product to be different (with about 67% of the respondents) & the product solves their issues (that the helmet addresses their safety concerns - with about 77% of respondents).

Therefore, this analysis shows that the product has good chance of capturing the two-wheeler market in India (subject to technological changes).

When the respondents were asked what is the price they were willing to pay for the product, majority of them were ready to pay for it in the range of Rs. 3000 to Rs. 5000 even if their current spending on helmets is between Rs. 1000 to Rs. 3000.

Final Analysis of Responses: After analyzing the various responses, it was found that even when people were having their current helmets, they were still interested in buying Guardian X. This suggests that based on the sample responses, it is hoped that introducing the product in the market will invite sales figures. Therefore, the final call taken is that the product can be launched (subject to test marketing).

Demand Forecasting

TOTAL AVAILABLE MARKET

Approximately, in India, there are 21 crores registered two wheelers, out of which close to 70% are using helmets. With an expectation of at least 40% will adopt the usage of the smart helmet, it is expected that close to 5 crore people will purchase Guardian X

Market Share

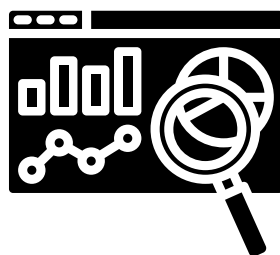
There are plenty of helmet manufacturers in India, such as Steelbird, Studds, Vega, Royal Enfield etc. But none of these have widespread dominance and are offering safety features such as impact detection, GPS tracking etc. Though there are some helmets in the 'smart helmets' in the market, it is restricted to providing only built in speakers and mic.

Assuming an optimistic expectation of penetrating the Safety Gear industry of **5%**, with a growth of **15%** further on.

Assuming a pessimistic situation, an expected 2% of the market is expected to be penetrated.

Therefore, putting in number terms, in the 1st year, it is expected to have a market share of about (5 % of 5 crores) **25 lakhs units**

Else, it is expected that in the 1st year (2% of 5 crores) to have a market share of 10 lakh units



Market Segmentation



Demographic

- **Age:** Primarily to middle-aged riders (18-45 years), as this group represents the largest two-wheeler riders in India.
 - **18-25:** Young, Tech Savvy and open to adopting innovative technologies.
 - **26-45:** Working professionals who often travel a lot, requiring safe rides.
- **Gender:** Due to increasing female two-wheeler riders, females as well as males are considered to be primary segment. An increasing proportion of the other gender will also be considered.
- **Income Level:** Middle - Upper Middle class will be considered as primary segment. This is because they can afford a premium product and are likely to value safety and comfort.
- **Occupation:** Based on occupation, for Guardian X, individuals who commute using two-wheelers and tech-savvy college students



Psychographic

- **Lifestyle:**
 - **Tech-Enthusiasts:** People who love riding as well as technology into their everyday tasks, drawn to increasing usage of IoT, GPS etc.
 - **Health & Safety Conscious Individuals:** Riders who prioritize safety during their thrilling rides, especially during their long-distance and urban travels.
 - **Urban Professionals:** Busy individuals who value safety and want convenience in their rides looking for navigation, communication & safety.
- **Values & Beliefs**
 - Riders who believe protecting themselves is important and are ready to invest in their future.
 - Consumers who want to showcase themselves as being in the top of the chain when it comes adopting new technology, symbolizing higher status in the society.



Geography

- The helmet is focused to be sold predominantly in urban areas with high traffic density with high two-wheeler vehicles usage.
- The helmet is primarily targeted in the Indian Market due to the high population with two-wheeler registrations and to address the need of road fatalities.
- The helmet is also targeted to be sold in areas of extreme weathers offering a added safety features to minimize the risk of accidents leading to deaths and injuries.

Behavioral

- **Usage Rate:**
 - **Heavy Users:** Daily commuters, adventure riders, people who are mainly dependent on two-wheelers, who prioritize on their safety gear with durability, comfort.
- **Benefits Sought:**
 - **Safety:** Riders who focus on safety in their rides, wanting an easy and added feature of protective features
 - **Convenience & Technology:** People who want a stop-free riding experience, requiring hands-free communication, GPS navigation etc.
- **Brand Loyalty:**
 - **Innovators & Early Adopters:** The customers who are more likely to try out new products like Guardian X are primary target, if they have a flavor for trying IoT based smart devices in their everyday rides.
 - **Price Sensitive Customers:** Customers who are price-conscious can be persuaded to upgrade to Guardian X to save on their recurring expenditure of helmet purchase by offering them a more durable and advanced version of helmet

Report by Team 7

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